Service Bulletin

Service Bulletin Number: 3666191	Released Date: 10-may-2016
Cummins®-Branded Delco-Remy™ Starters and Alternators	

Cummins®-Branded Delco-Remy™ Starters and Alternators

This Service Bulletin updates the release of Cummins-Branded Delco Remy Starters and Alternators for the B, C, L, M, N, K, and Q Series engine families. These products are identified with a Cummins label and are covered under New Engine Warranty on all applications except for the U.S./Canada recreational vehicle, school bus, fire truck and crash truck where the coverage will be limited to 2 years. Complete Cummins Form 6299 in the Cummins Warranty Administration Manual for more coverage information. The Cummins-branded starters and alternators will be identified by a data tag that states “Made for Cummins Engine Company.” These starters and alternators are serviced worldwide by the Cummins Authorized Repair Facilities. Refer to a Cummins® Authorized Repair Location for repair kits, kit content, and cross reference details.

BULLETIN CONTENTS

Section 1 - System and Component Description

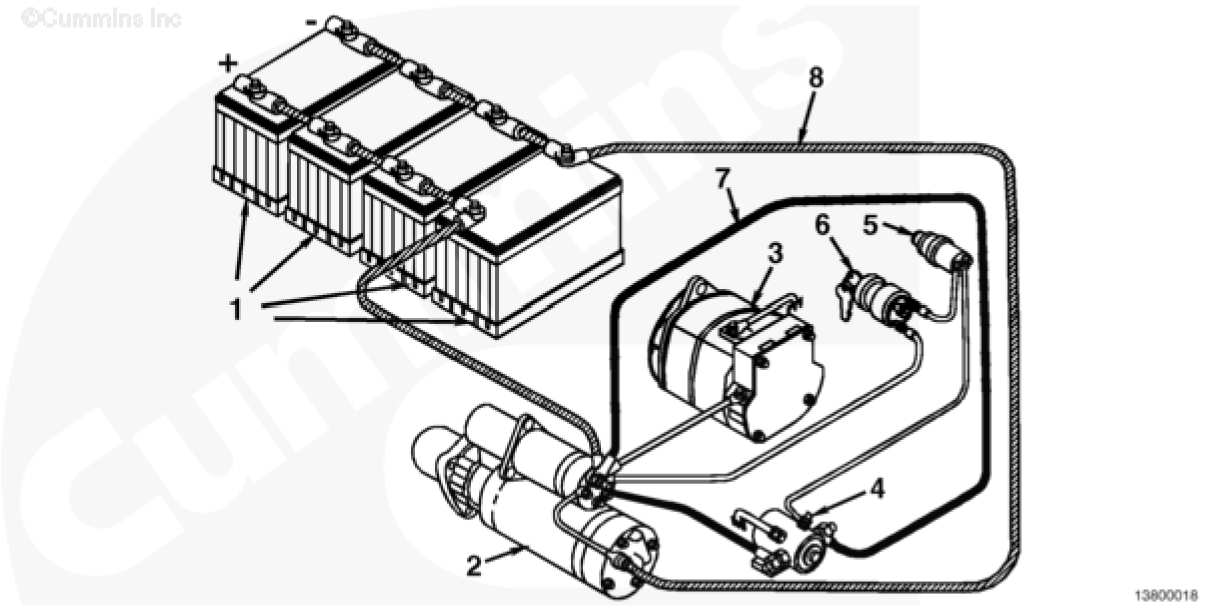


Table 1, Typical Heavy-Duty Electrical System

The heavy-duty electrical system consists of batteries, an alternator (generator), starter, control circuit, and wiring (Table 1).

1. Batteries
2. Starter
3. Alternator (generator)
4. Magnetic switch
5. Push-button switch
6. Keyswitch
7. Control, or relay, circuit wiring
8. Battery cables or cranking circuit

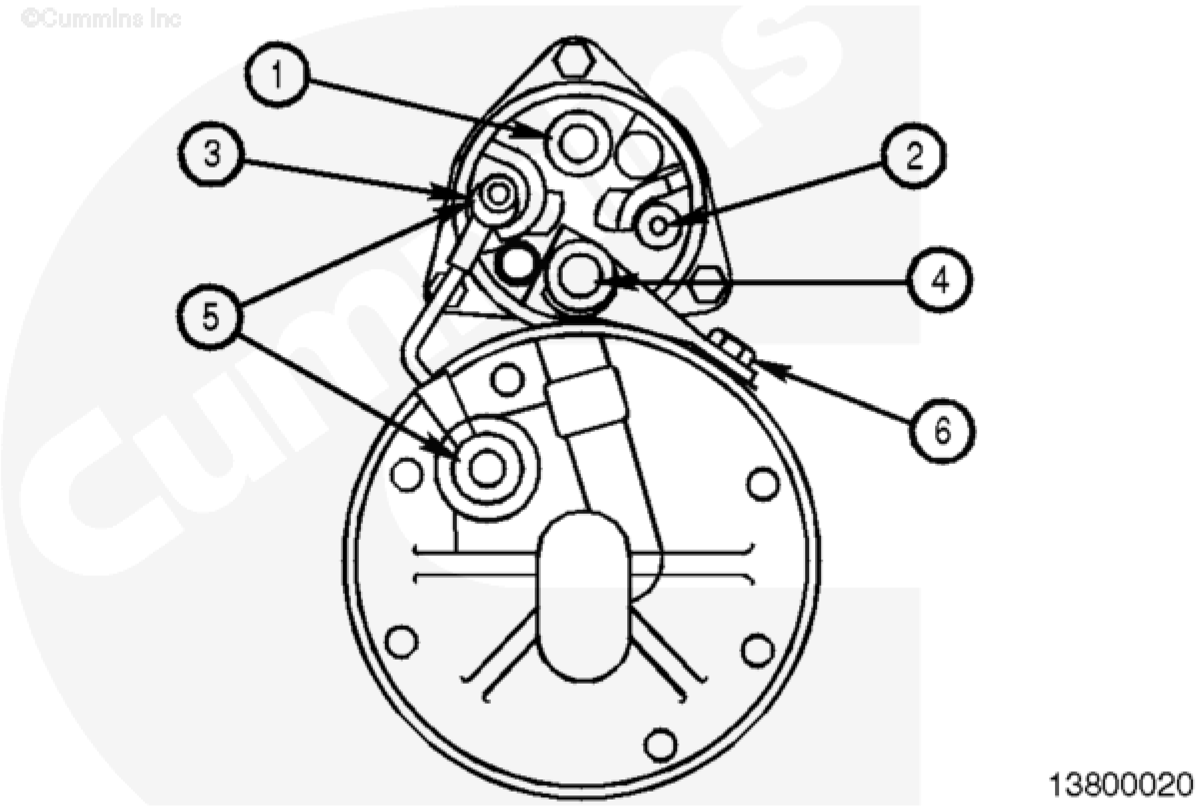


Table 2, Typical Starting Motor		
Key	**Terminal**	**Connected To:**
1	B	Battery
2	"S"	Magnetic switch
3	Ground strap[1]	Ground to motor and battery
4	Motor	Motor (no customer hookup required)
5	Ground[1]	Ground return terminals
6	"F"[1]	Field upfit terminal

1. **Not** all starters have this feature.

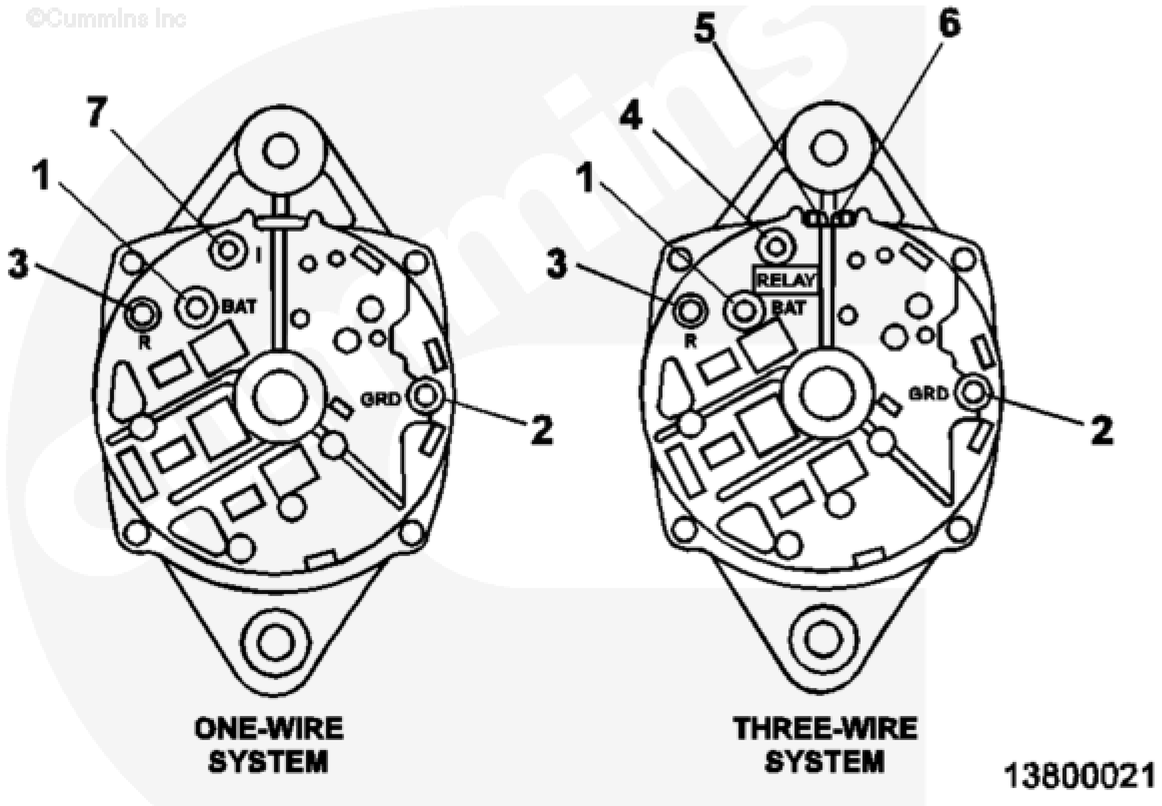


Table 3, Typical Alternator (Delco-Remy)		
Key	**Terminal**	**Connected To:**
1	Output or B+	Battery
2	Ground[1]or B-	Ground
3 and 4	Relay[1]	Charge indicator, automatic lock-out system, tachometer[2]
5	Terminal Number 1[1]	Indicator light
6	Terminal Number 2[1]	Voltage sensor
7	Indicator	Indicator light

1. **Not** all starters have this feature.
2. Relay terminal provides alternating voltage signal at nominal 1/2 system voltage (4 amps maximum draw).

Table 4, Alternator Frequency Factor	
Alternator Model	**Frequency Factor**
19SI, 21SI, 22SI, 23SI	0.100
20SI	0.117
26SI, 30SI, 33SI, 34SI	0.133

Section 2 - Operating Guidelines

⚠ CAUTION ⚠

To prevent damage to the starter, do not engage the starting motor for more than 30 seconds at a time. Wait 2 minutes between each attempt to start (electrical starting motors only).

If the engine does **not** start after three attempts, check the fuel supply system.

Absence of blue or white exhaust smoke during cranking indicates that fuel is **not** being delivered. The starter **must not** be used to prime the fuel system. Reference the appropriate Troubleshooting and Repair Manual for correct priming procedures.

The starter is an intermittent-duty electric motor that is capable of high output for short periods of time. Excessive cranking of the starting motor will lead to an unwarrantable failure.

Section 3 - Tool Requirements and Service Tools

This section outlines the tools required to troubleshoot and diagnose starter and alternator failures. It is divided into four categories. The first category is for tools that are required for on-engine diagnostics. The second category is for over-the-counter exchanges and rebuilding. The third category is for service tools. The fourth category is optional tools for servicing starters and alternators.

Table 5, Category 1 - Required On-Engine Diagnostics

Description	Service Tool Part Number
Multimeter	3377161
Clamp-on current probe	3823574
Digital optical tachometer	3377462

Table 6, Category 2 - Over-the-Counter Exchange or Rebuild

Description	Service Tool Part Number
Bench tester, three-phase 440 VAC, 50-Hz	3825047
Bench tester, three-phase 440 VAC, 60 Hz	3825048

Table 7, Category 3 - Service Tools

Description	Service Tool Part Number
T-wrench	3824866
T-wrench	3824867
Timing key	3824868

Table 7, Category 3 - Service Tools	
Description	**Service Tool Part Number**
Timing gauge	3824869
Abutment gauge	3824870
Abutment gauge	3825060
Abutment gauge	3825061
Timing gauge	3825062

Category 4 - Optional Tooling for Servicing Electrical System:

- Battery charger
- Armature growler.

General Information

Operating Temperatures

The operating range of the cranking motor and solenoid is -25 to 121°C [-13 to 250°F]. Military application starters have an operating range from -54 to 121°C [-65 to 250°F]. Locating the motor close to the exhaust manifold or other heat source can result in reduced motor performance with deterioration of insulation and rubber components; a minimum 2-inch clearance is recommended. Where relocation or rotation is **not** possible, metal shielding can effectively be used to protect the starter from radiated heat. Excessive heat will increase solenoid coil resistance and prevent starter engagement in hot conditions.

Wiring Guidelines for Proper Grounding

Establish proper routing. Avoid routing a taut cable between connections. When routing battery cables to the cranking motor, anchor the cables to the engine, transmission, or cranking motor, and allow for relative movement. It is highly recommended that a ground cable from the starter ground terminal to the battery negative (-) terminal be utilized and the frame **not** be relied upon for ground return in the starting circuit, which will prevent corrosion from creating excessive voltage drops. If a ground terminal is **not** available, use the starter mounting ears. Using a ground cable is especially critical in applications utilizing a wet flywheel housing, where reliance on frame grounding can create a current path through the crankshaft and main bearings. Limit the number of cables attached to the battery terminal of the solenoid to no more than two.

⚠ CAUTION ⚠

Never use any type of internally or externally toothed lock washer (also called star washer), on wiring connections. The points of the star washer cause current to be conducted only through the points and lead to corrosion and a high-resistance electrical connection. Use a standard flat washer with a larger flat surface to prevent this problem.

Starter Grounding

There have been several installations identified over the last several years in which engine crankshaft and bearing damage occurred because of inadequate engine/starting motor grounding. While these issues occurred on B and C Series engines, it is possible for them to occur on other Cummins Inc. engines. A complete electrical flow path that starts at the battery negative (-) terminal **must** be provided in all installations. If **not**, the electrical current will seek its own path to ground that can cause engine damage. The proper way to ground a starting motor depends on whether the starter is insulated or non-insulated, and whether the starter mounting option is intended for a “wet” or “dry” flywheel housing.

Starting Motors: Insulated Versus Non-insulated

Insulated starters have a special grounding stud called the ground lug, usually located at the rear of the starter housing. This the **only** acceptable grounding connection on this type of starter. The starting motor housing is electrically insulated from the motor inside. Because of this, insulated starters can **not** be grounded through the engine by face-to-face contact between the starter and the flywheel housing. The starter and flywheel housing can **not** be grounded by attaching a ground strap to the mounting ear of the starter. The **only** way this type of starter can be grounded is by attaching a ground cable or strap to the ground lug and completing the circuit to ground. The engine needs to have a separate ground strap or cable to be sure that the engine block is effectively grounded. This is especially true if the engine is soft-mounted on rubber vibration isolators.

Non-insulated starters can be grounded in several ways. Since the starting motor housing and mounting ears are electrically connected to the starter motor inside, the starter can be grounded by attaching a cable or strap to any of the mounting ears on the starter and completing the circuit to ground, or by grounding through the engine by the physical face-to-face connection of the starter housing to the flywheel housing. It is very important to note that the starter can be grounded through the engine **only** if a dry starter mounting (no gasket or sealant) is used and the engine has a separate ground cable attached to the block. Sometimes a non-insulated starter will have a grounding stud, the same as an insulated starter. This stud can be used as an acceptable way to ground either type of starting motor.

Starter Motor Mountings: Wet Versus Dry

The type of starter mounting used matters **only** on non-insulated starters. If the starter mounting option selected is designed for a wet flywheel housing where the flywheel housing contains lubricating oil, a sealing gasket will be installed between the starter and flywheel housing mounting surface. The starter capscrews can be coated with sealant. Because these sealing items can insulate the starter housing from the flywheel housing, the **only** acceptable way to ground the starter is by attaching a ground strap or cable to the starter mounting ears or, if present, the starter ground lug. If a ground strap or cable is **not** utilized, the starting motor can seek a ground through the internal components of the engine. This can result in progressive main bearing and crankshaft damage in the form of electrically induced surface pitting and/or eventual component failure.

When using a dry flywheel housing, the starter motor can be grounded through any of the methods described previously (as determined by the type of starter), as long as there is no gasket or sealant included in the mounting option, and the engine itself has a separate ground cable or strap attached to the block.

	Table 8, Starter (ST) Options	
	Insulated (wet)	**Non-insulated (dry)**
Wet Flywheel Housing Option	Ground starter by attaching a ground cable or strap to the lug and completing the circuit to ground. Use a wet starter mounting option.	Ground the starter by attaching a cable or strap from the battery negative (-) terminal to the starter ground lug. If the ground lug is **not** present, use the mounting ears on the starter. Use a wet starter mounting option.
Dry Flywheel Housing Option	Ground starter by attaching a ground cable to the ground lug and completing the circuit to ground. Use either a wet or dry starter mounting option.	Ground the starter by attaching a cable or strap from the battery negative (-) terminal to the starter ground lug. If the starter ground lug is **not** present, use the mounting ears on the starter. Either use wet or dry starter mounting option, or ground through the engine by physical face-to-face connection of the starter housing to the flywheel housing. Use **only** with the dry starter mounting option.

Solenoid Orientation

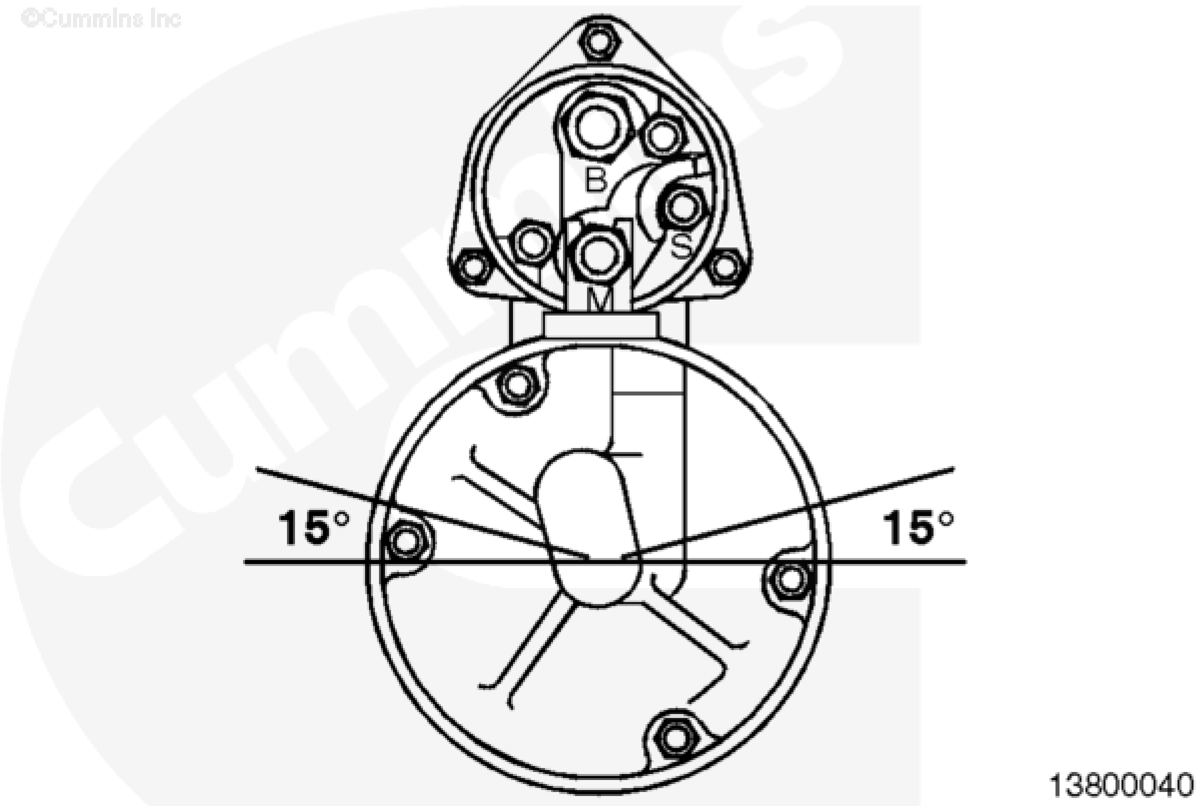


Fig. 1, Starting Motor

When positioning the motor at the mounting location, it is desirable to place the solenoid above the centerline of the motor (preferably a minimum of 15 degrees above the horizontal). This will prevent potential accumulation of foreign material or moisture that can restrict operation in the tower portion of the shift lever housing. It also reduces the possibility of road debris being thrown up and breaking the solenoid cap.

Mounting Specifications

1. The flange-to-flywheel distance is the distance on the engine from the cranking motor mounting surface to the ring gear face. This distance is 50.8 ±1.52 mm [2.0 ±0.060 in]. This dimension **must** include spacers or gaskets used on the engine. The cranking motor is designed to function within this dimension; if it is altered, pinion or ring gear damage can occur.
2. Delco Remy America recommends the use of 12-point, flanged-head capscrews.
3. No lock washers, star washers, or soft washers, including copper electrical connections, can be used. Lock washers and star washers do **not** have a rectangular cross section and can result in a loss of clamp force on the mounting capscrews. The yield strength of soft washers or copper connections can be exceeded with mounting capscrews tightened. In each case, a loose starter mounting will result.

Cranking Motor Operation

⚠ CAUTION ⚠

If the engine does not start and must be cranked again, always wait 2 to 3 seconds to allow the pinion and ring gear to stop. If the starter is reengaged immediately, the pinion can be milled or the armature twisted. The starter is intended only for cranking the engine and must not be used to “bump” the engine.

If the engine does **not** start within 5 seconds of cranking, follow the cold start procedure in the corresponding Cummins operation and maintenance manual. Operating the cranking motor for more than 30 seconds at a time causes undue stress and/or severe internal heat buildup. A minimum 2-minute cooling period is necessary between extended cranking cycles.

Magnetic Switch Mounting

When mounting a magnetic switch on a vehicle, the axis of the plunger **must** be horizontal to the ground and perpendicular to the axis of travel of the vehicle. This is to make sure that the switch is **not** activated by the movement of the vehicle. Do **not** mount the switch on the engine or any metal that can possibly resonate as the result of road or engine vibration. For engines with separate fuel shutoff solenoids, Cummins recommends the use of two magnetic switches: One for the fuel shutoff circuit and one for the starting circuit. The fuel shutoff solenoid **must not** be wired to the S terminal of the starting motor. Wiring to the S terminal will cause the shutoff solenoid and/or starter to malfunction.

Cold Cranking

Reference the corresponding operation and maintenance manual for cold weather operation. A starting aid will possibly be required.

Solenoid Average Current Draw

- For 12-VDC systems, apply 10 VDC to the S terminal.
- For 24-VDC systems, apply 20 VDC to the S terminal.
- For some 32-VDC systems, apply 30 VDC to the S terminal.
- For some 32-VDC and all 64-VDC systems, apply 30 VDC to the B+ terminal.

Table 9, Solenoid Current Draw

Motor	Voltage	PI and HI Current	HI Current
28MT	12 VDC	69 Amps	13 Amps
28MT	24 VDC	120 Amps	13 Amps
37MT	12 VDC	74 Amps	19 Amps
37MT	24 VDC	36 Amps	6 Amps
41/42MT	12 VDC	97 Amps	18 Amps
41/42MT	24 VDC	57 Amps	13 Amps
50MT	12 VDC	86 Amps	15 Amps
50MT	24 VDC	49 Amps	6 Amps

Table 9, Solenoid Current Draw			
Motor	**Voltage**	**PI and HI Current**	**HI Current**
50MT	32 VDC	38 Amps	6 Amps
50MT	64 VDC	10 Amps	2 Amps

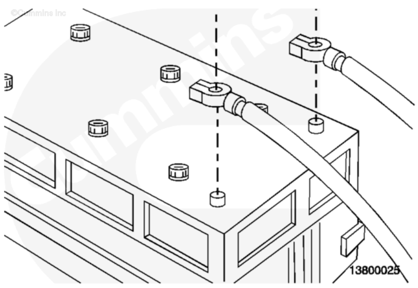
Section 4 - Troubleshooting and Repair

A. Starter Troubleshooting:

Table 10, Engine Will Not Crank or Cranks Slowly		
Cause		**Correction**
Battery charge low	---	Check battery (Step A-1). Charge battery as required.
OK		
Engine parasitic loads prevent engine from attaining minimum cranking speed.	---	Remove or disengage all engine parasitic loads. Reference the appropriate Troubleshooting and Repair Manual for correct engine cranking speed.
OK		
Loose or corroded starting circuit connections	---	Inspect and clean all connections between the starter and battery and the magnetic switch. Check and tighten all connections.
OK		
Poor starting motor ground	---	Check for proper ground between starting motor and battery negative terminal. Repair or replace as necessary.
OK		
Cranking speed is low due to cold ambient temperature	---	Use a starting aid and the correct oil grade. Reference the appropriate Troubleshooting and Repair Manual for oil specifications.
OK		

Table 10, Engine Will Not Crank or Cranks Slowly		
Cause		**Correction**
Engine rotation is restricted	---	Rotate the crankshaft, and remove parasitic loads or engine rotation restrictions.
OK		
Cranking circuit or battery cable voltage drop excessive	---	Check the voltage drop in the starting circuit (Step A-2). Repair or replace cables as required.
OK		
Control circuit or starter relay voltage drop excessive	---	Check the voltage drop in starter circuit (Step A-3). and repair or replace the starter relay as required.
OK		
Damaged starter and/or ring gear	---	Check the flange-to-flywheel dimension 50.8 mm ± 1.524 mm [2.0 in ± 0.06 in]. Remove the starter and inspect the pinion and ring gear for damage. Replace the gear or starter as required.
OK		
Starting motor failed	---	Replace starting motor.

Step A-1, Battery Check



⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

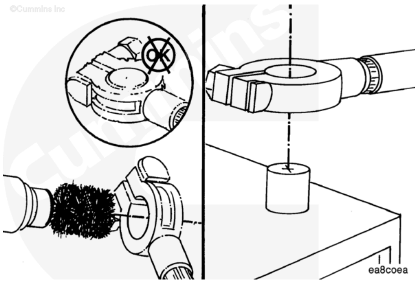
⚠ WARNING ⚠

Acid is extremely dangerous and can damage the machinery and can also cause serious burns. Always provide a tank of strong soda water as a neutralizing agent when servicing

batteries. Wear goggles and protective clothing to reduce the possibility of serious personal injury.

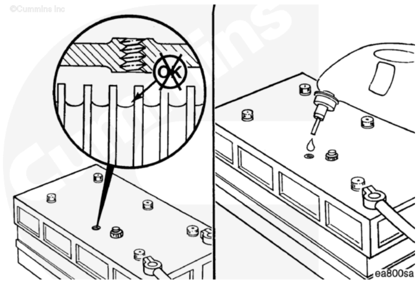
Label and disconnect all battery cables.

Clean corrosion and debris from battery and terminals.



Note : If water is added to the battery, the battery must be charged before any testing can be accomplished.

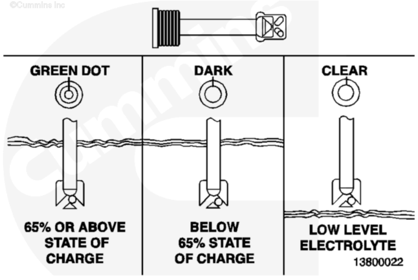
If conventional batteries (**not** maintenance-free batteries) are used, check the electrolyte level in each cell. If it is low, add distilled water to bring the level up to the top of all plates. Refer to the OEM specifications for the battery.



With conventional batteries, check the electrolyte specific gravity with a hydrometer. To check the specific gravity, the battery temperature will be 26.67°C [80°F]. All cells will **not** be less than 1.230. The difference between high and low readings will **not** exceed 0.050. If the readings indicate a difference of more than 0.050, replace the battery. If the readings indicate less than 0.050 difference but one or more cells indicate less than 1.230, recharge the battery.

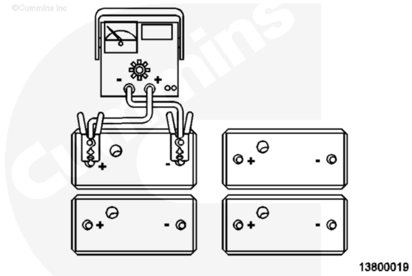
Battery Specific Gravity at 26.67°C [80°F]	
Specific Gravity	**State of Charge**
1.260 to 1.280	100 percent
1.230 to 1.250	75 percent
1.200 to 1.220	50 percent
1.170 to 1.190	25 percent
1.10 to 1.130	Discharged

Check the “eye” on the maintenance-free battery. Refer to the OEM specifications.

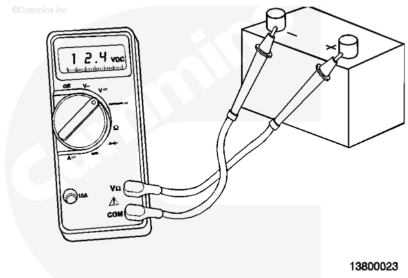


Note : For conventional batteries, remove the vent caps prior to removing surface charge. If a blue haze is observed in any cell while removing the surface charge, replace the battery. After the surface charge has been removed, install all vent caps.

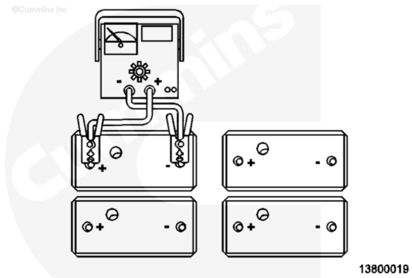
Remove the surface charge by attaching the battery to a 300-amp load for 15 seconds for heavy-duty batteries.



Remove the load and wait 1 minute; if the battery voltage is greater than or equal to 12.4-VDC, continue testing. If the voltage is below 12.4-VDC, recharge or replace the battery.

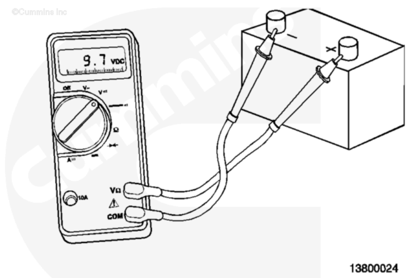


Load-test the batteries at 1/2 of the cold cranking amp rating of the battery (rating at -18°C [0°F]) for 15 seconds

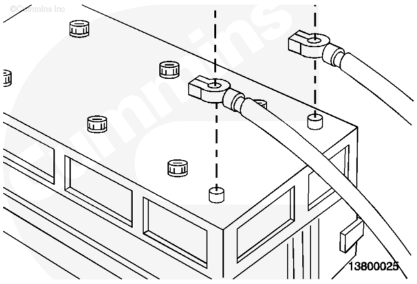


Check the battery voltage and compare to the table below:

Temperature and Voltage Relationship								
Temp [°F]	70	60	50	40	30	20	10	0
Temp (°C)	21	16	10	4	-1	-7	-12	-18



Temperature and Voltage Relationship								
Miimum Voltage (VDC)	9.6	9.5	9.4	9.3	9.1	8.9	8.7	8.5



⚠ WARNING ⚠

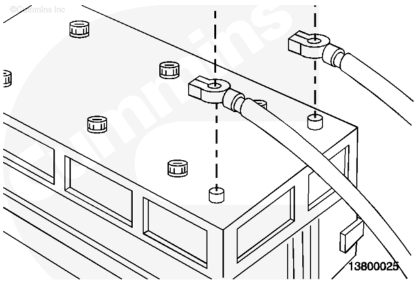
Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

⚠ WARNING ⚠

Acid is extremely dangerous and can damage the machinery and can also cause serious burns. Always provide a tank of strong soda water as a neutralizing agent when servicing batteries. Wear goggles and protective clothing to reduce the possibility of serious personal injury.

Connect all battery cables.

Step A-2, Battery Cable Test Procedure



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⚠ CAUTION ⚠

If the vehicle has a combination 12 and 24-VDC system using a series-parallel switch or a T/R alternator, do not use this procedure. Refer to a Cummins® Authorized Repair Location.

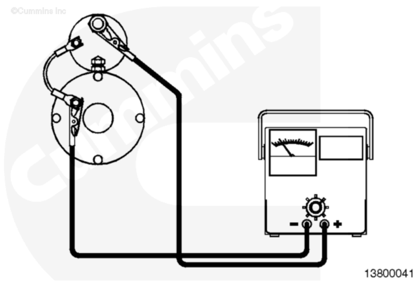
Note : For 24-VDC systems, use a 24-VDC carbon-pile tester. As an alternate, connect **only** one 12-VDC battery to the system. Disconnect all other batteries. Test at 12-VDC but use amperage specified for a 24-VDC system. Immediately upon completion of tests, reconnect batteries in the approved manner for 24-VDC system.

Disconnect negative (-) and positive (+) battery cables.

Note : Information for the following procedures has been supplied by Delco Remy America.

Connect the carbon-pile tester positive (+) lead to the starter solenoid "Bat" terminal.

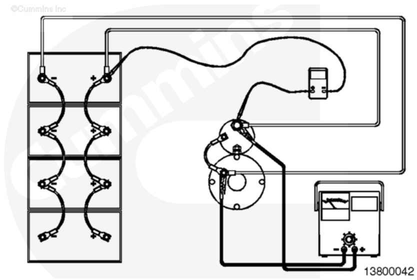
Connect the carbon-pile tester negative (-) lead to the starter solenoid ground terminal.



Note : Multimeter connection **must** be made to the starter solenoid terminal, **not** to the carbon-pile tester clamp.

Connect the multimeter positive (+) lead to the starter solenoid "Bat" terminal.

Connect the multimeter negative (-) lead to the positive (+) battery post.



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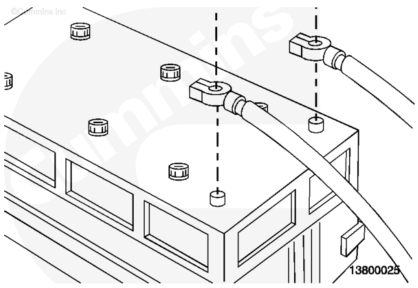

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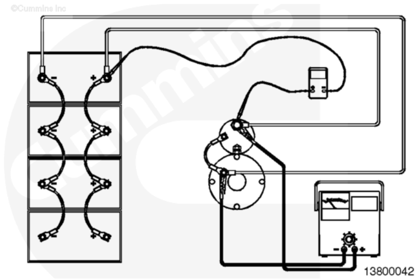
Connect negative (-) and positive (+) battery cables.

Turn on the carbon-pile tester and adjust to 500 amps for 12-VDC systems (250 amps for 24-VDC systems).

Record multimeter display. This value is the cable positive (+) voltage drop (V1).



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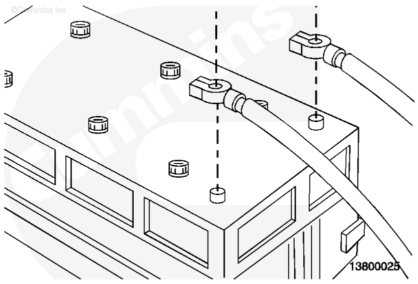

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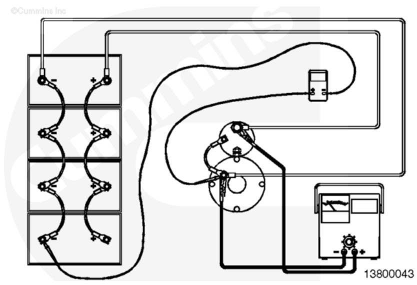
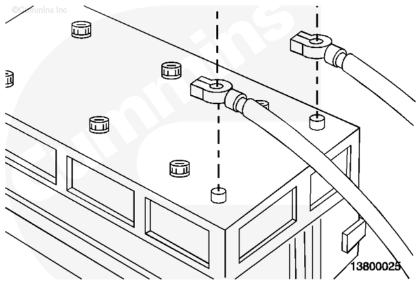
Disconnect negative (-) and positive (+) battery cables.

Disconnect multimeter from the previous measurement.

Connect the multimeter positive (+) lead to the starter ground terminal.

Connect the multimeter negative (-) lead to the negative (-) battery post.





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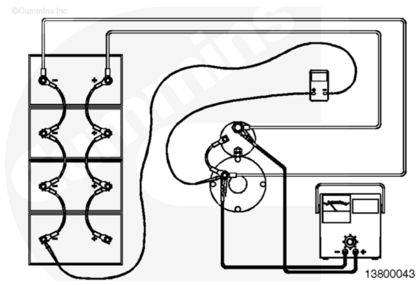
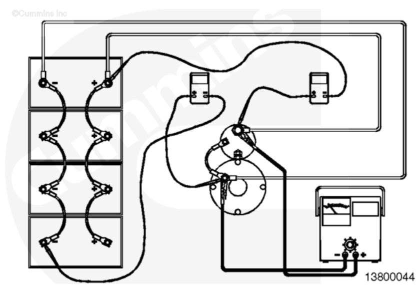
Connect negative (-) and positive (+) battery cables.



Turn on the carbon-pile tester and adjust to 500 amps for 12-VDC systems (250 amps for 24-VDC systems).

Record the multimeter display. This value is the cable negative (-) voltage loss (V2).

Disconnect all meters and return system to approved operating configuration.



Add the positive (+) cable voltage loss and the negative (-) cable voltage loss (V1+V2=V3). The sum of the two equals total cable voltage loss (V3). Reference the table below for maximum cable voltage loss.

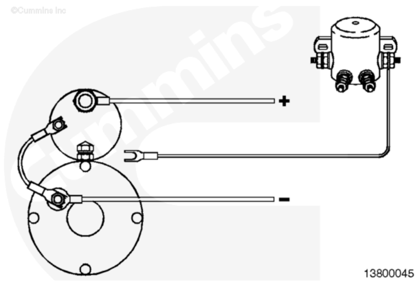
Maximum Cable Voltage Loss		
Starter	**12-VDC Value**	**24-VDC Value**
37MT, 40MT, 41MT, 42MT	0.500 VDC	1.000 VDC
50MT	0.400 VDC	1.000 VDC

Step A-3, Starter Solenoid Circuit Test

Note : For starters with integral magnetic switch, use test procedures specific to those starters.

Note : When testing 24-VDC systems, use the same temporary 12-VDC connection that was outlined in the Battery Cable Test (Step A-2).

Disconnect the “S” lead from the starter solenoid.

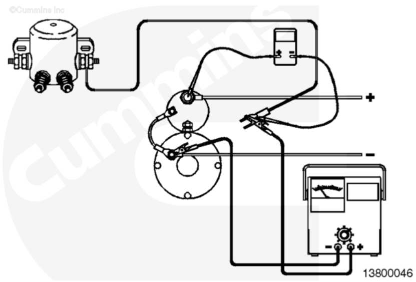


Connect the carbon-pile tester positive (+) lead to the “S” lead that was disconnected in the previous step (a small clamp or jumper wire can be used).

Connect the carbon-pile tester negative (-) lead to the starter ground terminal.

Connect the multimeter positive (+) lead to the “Bat” terminal on the starter solenoid.

Connect the multimeter negative (-) lead to the carbon-pile positive (+) and the starter solenoid “S” leads previously connected.



Note : If the magnetic switch does not close on a 12-VDC vehicle, perform the magnetic switch circuit test outlined in this bulletin.

Note : In the following procedure, on a 24-VDC vehicle, if the temporary 12 VDC will not close the magnetic switch, bypass it with a heavy jumper connected between the two large studs on the magnetic switch. Electrically, this does the same thing as pushing the button and closing the switch. With the button bypassed, the jumper must be disconnected after each voltage reading.

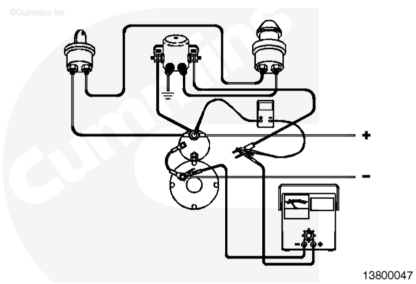
With the key on, push the starter button. Listen for the sound of the magnetic switch closing.

The multimeter display will be 0 VDC.

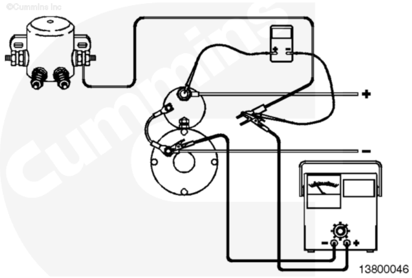
Turn on and adjust the carbon-pile tester to 100-amp load (60 amps for 24 VDC).

The multimeter display can **not** exceed 1.0 VDC for 12-VDC system, 2.0 VDC for a 24-VDC system.

Note : If circuit voltage is less than the above maximum, solenoid circuit is good. If voltage exceeds maximum, reference the Starter Solenoid Wiring Test in this bulletin.



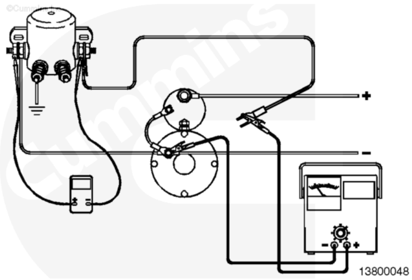
Step A-4, Starter Solenoid Wiring Test



Preparatory Step

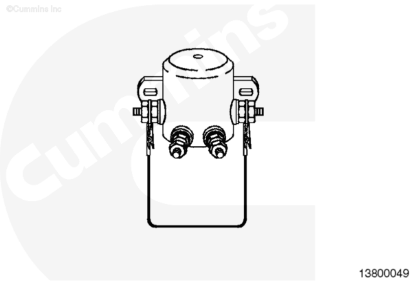
- Disconnect “S” lead from starter.
- Connect carbon-pile tester.

Reference the Solenoid Circuit Test for proper connection procedure for the carbon-pile tester in this service bulletin.



Connect the multimeter positive (+) lead to the magnetic contactor “Bat” terminal.

Connect the multimeter negative (-) lead to the magnetic contactor “S” terminal. If voltage is present, switch the multimeter leads around.



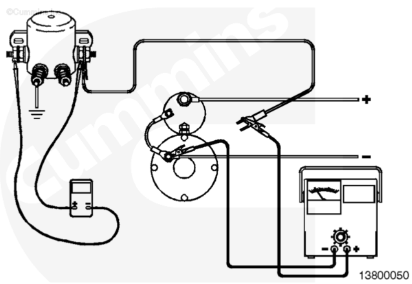
Note : In the following step, on a vehicle with a 24-VDC system, if the temporary 12-VDC will not close the magnetic switch, bypass it with a heavy jumper connected between the two large studs on the magnetic switch. Electrically, this does the same thing as pushing the button and closing the switch. With the button bypassed, the jumper must be disconnected after each voltage reading.

With the keyswitch on, push the starter button.

Turn on the carbon-pile tester and adjust to 100-amp load (60 amps for a 24-VDC system).

Record the multimeter display voltage (V1).

Disconnect the multimeter



Connect the multimeter positive (+) lead to the magnetic contactor “S” lead and the carbon-pile tester positive (+) lead.

Connect the multimeter negative (-) lead to the magnetic contractor “Bat” lead.

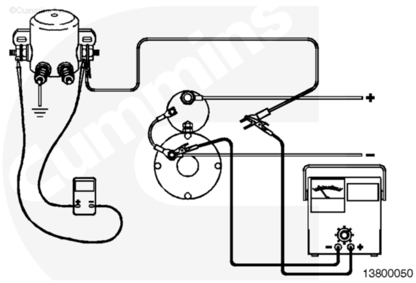
With the keyswitch on, push the starter button.

Turn on the carbon-pile tester and adjust to 100 amps for 12-VDC system (60 amps for 24-VDC system).

Record the voltage displayed on the multimeter (V2).

Add voltage loss (V1) to voltage loss (V2); the sum is total voltage loss (V1+V2=V3). Reference the table below for maximum voltage loss.

Maximum Voltage Loss	
12-VDC System	**24-VDC System**
0.8-VDC	1.8-VDC



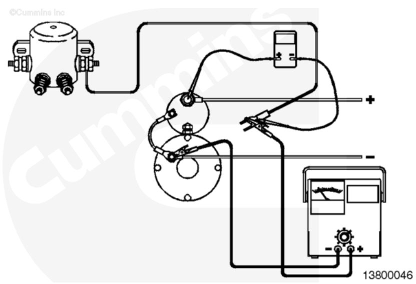
Step A-5, Magnetic Switch Contactor Test

Preparatory Step

- Disconnect “S” lead from the starter.
- Connect carbon-pile tester.

Note : Perform this test only if the magnetic switch closed in the Starter Solenoid Circuit Test contained in this bulletin.

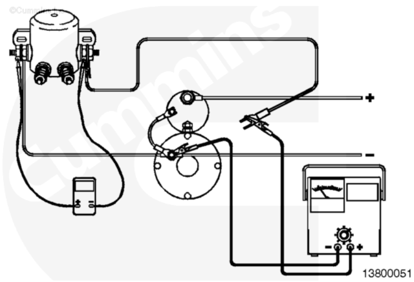
Reference the Starter Solenoid Circuit Test in this service bulletin for correct connections for the carbon-pile tester.



Connect the multimeter positive (+) lead to the magnetic contactor “Bat” post.

Connect the multimeter negative (-) lead to the “S” post of the magnetic contactor.

Battery voltage will show on multimeter display.



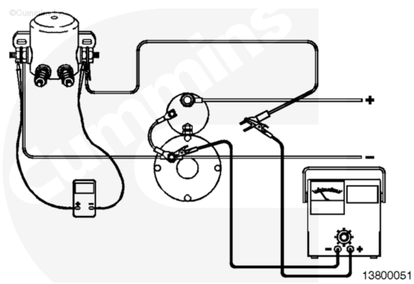
With the keyswitch in the ON position, push the start button. The multimeter will display 0-VDC.

Turn and adjust the carbon-pile tester to a 100-amp load for 12-VDC systems (60 amps for 24-VDC systems).

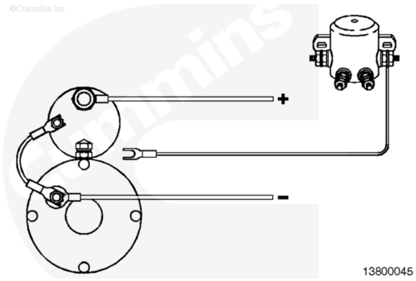
Record the voltage displayed on the multimeter.

Magnetic switch voltage loss **must not** exceed 0.2-VDC for 12- and 24-VDC systems.

Replace magnetic contactor if voltage loss is excessive.



Step A-6, Magnetic Switch Circuit Test

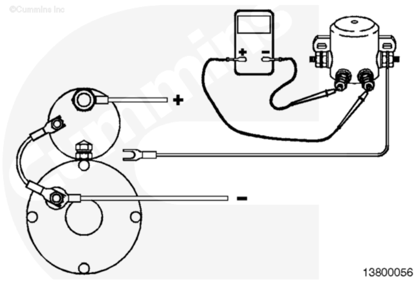


Note : For motors that include the magnetic switch as part of the motor (42MT IMS), reference the Delco Field Service Bulletin DR77.

Note : This test must be performed using full voltage of the system.

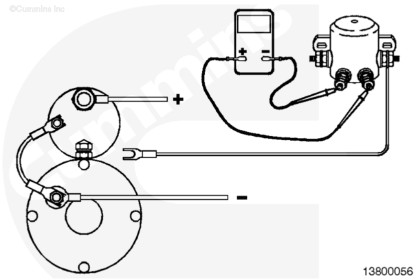
Note : High resistance or an open circuit can cause a magnetic switch to open prematurely or prevent it from closing.

Disconnect the "S" lead.



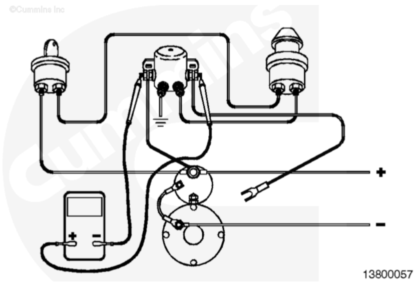
Connect the multimeter negative (-) lead to the magnetic switch ground terminal (one of two small terminals).

Connect the multimeter positive (+) lead to the remaining terminal (positive [+] supply from the push button).



With the key on, push the start button and listen for the click of the magnetic switch closing.

Record the voltage on the multimeter display. The voltage will be system voltage.



Perform this test if the magnetic switch closed.

Connect the multimeter positive (+) lead to the magnetic switch "S" post (large post).

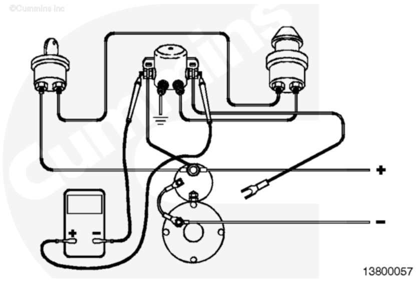
Connect the multimeter negative (-) lead to the magnetic switch "Bat" post (large post).

With the keyswitch on, push the start button. Record the voltage. If the voltage is within 1-VDC for 12-VDC system and 2-VDC for 24-VDC system below battery voltage, the circuit is operational.

Perform this test if the magnetic switch did **not** close.

Connect the multimeter leads to the same location as the previous step.

If the voltage readings are within the tolerances listed in the previous step, replace the magnetic switch and retest the circuit.



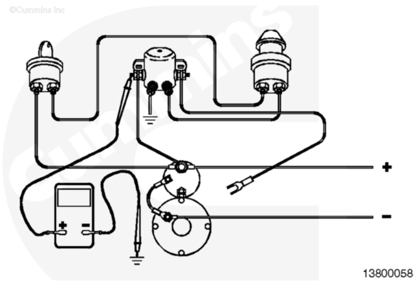
Perform this test if the voltage is more than 1-VDC for a 12-VDC system and 2-VDC for a 24-VDC system below battery voltage.

Connect the multimeter positive (+) lead to the “Bat” post of the magnetic switch.

Connect the multimeter negative (-) lead to chassis ground.

With the keyswitch on, push the start button. If the voltage is within the above specification, replace or repair ground lead at the magnetic switch.

If the voltage is **not** within the above specification, replace the ground lead at the chassis ground.

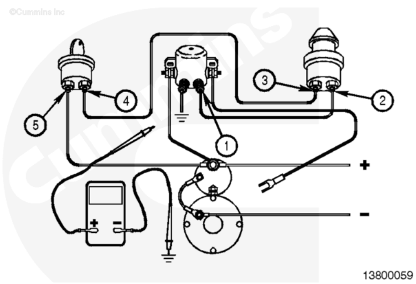


Connect the multimeter negative (-) lead to chassis ground.

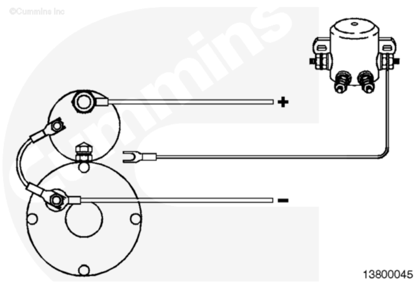
With the keyswitch on, push the start button and connect the multimeter positive (+) lead to the following locations:

1. Wire between push button and magnetic switch
2. Push button
3. Wire between push button and keyswitch
4. Keyswitch
5. Wire between keyswitch and solenoid “Bat” terminal.

If the voltage is **not** within 1-VDC for a 12-VDC system and 2-VDC for a 24-VDC, replace wire or component and retest the faulty area.

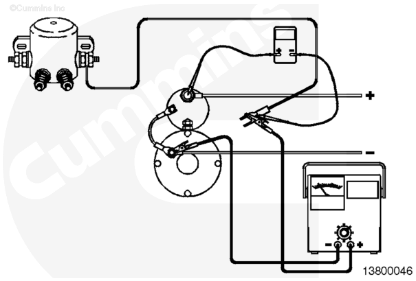


When all tests have been completed, disconnect the multimeter and reconnect the “S” wire to the starter solenoid.



Step A-7, Starter Replacement Determination

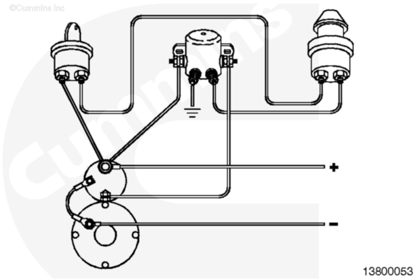
The following are additional checks that can be made before considering starter replacement, and the criteria for starter replacement if necessary.



Step A-8, Cold Weather Cranking

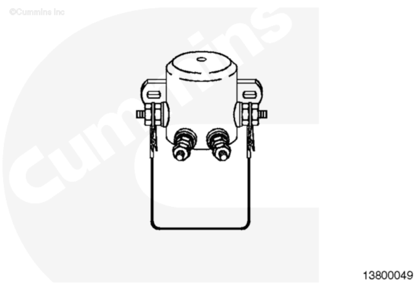
⚠ CAUTION ⚠

Large terminal studs on the magnetic switch are at battery voltage. Engine can crank when jumper is connected.



Starter circuits with a magnetic switch can also fail to hold in during cold weather starting and low voltage, even though the switches and circuits tested OK at higher temperatures. This condition can sound as though the starter is failing to stay engaged with the engine. It is caused by the cold weather low voltage of the system releasing the electrical connection of the magnetic switch. The following steps are for testing for this condition.

With the keyswitch in the ON position, press the start button and have an assistant clamp a heavy-duty battery jumper cable between the two large studs on the magnetic switch.



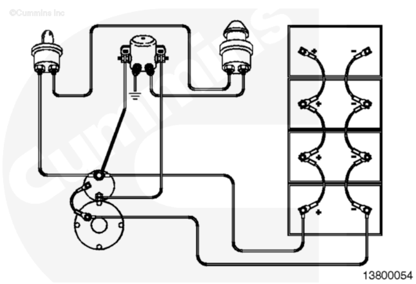
Immediately remove the jumper cable to stop engine cranking.

If engine starts with jumper, replace the magnetic switch.

After replacing switch, check the engine for proper starting operation. Make sure the starter mounting bolts are tight.

Step A-9, Available Cranking Voltage

If batteries, switches, and wiring have been checked and starter still cranks slowly, check for available voltage when cranking the engine.

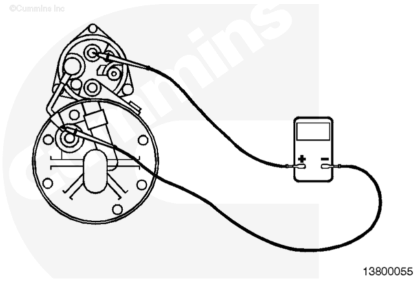


Connect multimeter positive (+) lead to starter solenoid “Bat” terminal.

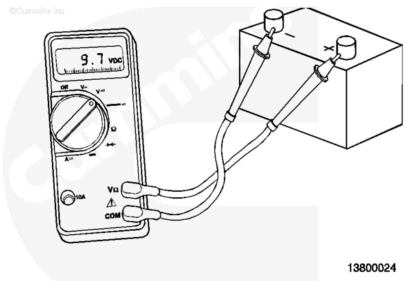
Connect multimeter negative (-) lead to the starter ground terminal.

With keyswitch on, press start button.

Check multimeter display. If voltage is less than 9.0 VDC for 12-VDC system (18 VDC for 24-VDC systems) when cranking, check the battery interconnecting cables as specified in the next step.

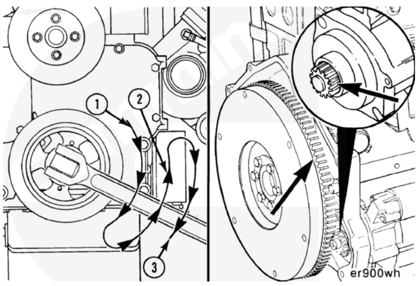


Measure voltage across each battery while cranking. Touch multimeter leads to terminals of every battery while cranking. If the difference between any two battery readings in the same box is more than 0.5- VDC or any cable or connection feels warm to the touch, check or replace any interconnecting cable(s) as required.



Step A-10, Ring Gear and Pinion Check

Inspect the pinion and ring gear while assistant bars the engine over. Make sure to check the entire ring gear.If pinion is damaged, replace the starter. If ring gear is damaged, replace the ring gear. A damaged ring gear can mean a damaged pinion.



Alternator General Information

Operating Temperatures

All Delco Remy America alternators are rated for inlet air temperatures of -35 to 93°C [-31 to 199°F], measured 25.4 mm [1-inch] from the rear of the unit, except for the 26SI. The 26SI has a maximum rating of 85°C [185°F] because of its sealed electronics. Avoid mounting near any heat sources, such as manifolds, turbochargers, and so forth. Adequate clearance **must** also be provided at the rear of the alternator for cooling air to enter (the alternator fan pulls air through the front of the unit).

Environmental Protection

The second most common cause of failure of Cummins Inc. alternators is contamination by dirt, chemicals, water, or unknown substances. While the alternator is laboratory tested to withstand salt spray, dust, and high humidity, it is necessary to mount the alternator to minimize environmental and application contamination. Exposure to salt spray over an extended period will shorten alternator life, as well as increase circuit resistance by causing corrosion of terminal connections for all electrical components. Exposure to engine oil or oil mist from engine breather tubes can cause brush problems in brush alternators, as well as attract other debris. Any contamination of the rear of the alternator can reduce airflow, which will eventually overheat the diodes.

Wiring Guidelines and Proper Grounding

⚠ CAUTION ⚠

Never use any type of internally or externally toothed lock washer (also called star washer), on wiring connections. The points of the star washer cause current to be conducted only through the points and lead to corrosion and a high-resistance electrical connection. Use a standard flat washer with a larger flat surface to prevent this problem.

Establish proper cable routing, and determine cable lengths. Avoid routing taut cables between connections. The maximum voltage drop in the charging system at rated alternator output is 0.5-VDC for a 12-VDC system, 1.0-VDC for a 24-VDC system, and 1.5 VDC for a 32-VDC system. The voltage drop in the charging system **must** be kept to a minimum in order to make sure enough voltage exists to charge the batteries.

While all Delco Remy America alternators are negative grounds, it is suggested that a full copper return be provided to the negative (-) terminal of the batteries. This also will make sure that as the vehicle ages, corrosion of the frame and ground strap does **not** cause excessive voltage drops. Also, painted mounting brackets can prevent a complete electrical circuit. Be sure all electrical connections, including ground connections, are metal-to-metal connections.

Mounting Hardware

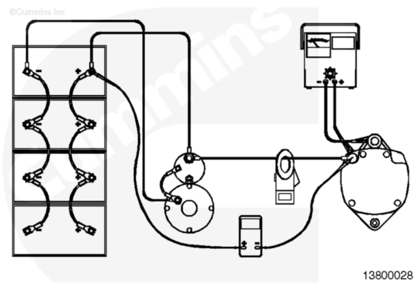
1. All capscrews **must** be a minimum of 1/2 inch.
2. **Always** use a hardened steel washer under mounting bolt heads, or use flanged-head hardware. The washer **must** match the bolt's diameter.
3. **Never** use any type of lock washer, including toothed or split lock washers, on any mounting connection. Lock washers are **not** recommended for mounting bolts because their trapezoidal cross section provides less clamping surface and higher clamping stress.

B. Alternator Troubleshooting

Alternator Not Charging, Insufficient Charging, or Overcharging		
Cause		**Correction**
Battery or wiring connections are loose, broken, or damaged	---	Repair wiring; clean and tighten connections.
OK		
Alternator belt is slipping, damaged, or missing	---	Check the belt for damage and correct tension. Refer to the appropriate OEM service manuals.
OK		

Alternator Not Charging, Insufficient Charging, or Overcharging		
Cause		**Correction**
Alternator seized	---	Replace the alternator.
	OK	
Alternator pulley is loose on the drive shaft	---	Repair or tighten the pulley.
	OK	
Connection to ground is poor	---	Inspect the alternator mounting hardware for a good electrical connection to the battery. Remove paint or debris from the ground connection.
	OK	
Voltage drops in charging circuit are excessive	---	Check voltage drops (Step B-1).
	OK	
Alternator voltage is excessive	---	Check the alternator voltage (Step B-2).
	OK	
Alternator is **not** performing to rated performance	---	Check the alternator performance (Step B-3).

Step B-1, Alternator Wiring Test Procedure



⚠ WARNING ⚠

Batteries can emit explosive gases. To avoid personal injury, always ventilate the compartment before servicing the batteries. To avoid arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

⚠ WARNING ⚠

Acid is extremely dangerous and can damage the machinery and can also cause serious burns. Always provide a tank of strong soda water as a neutralizing agent when servicing batteries. Wear goggles and protective clothing to avoid serious personal injury.

Note : Systems that are 24 VDC must be connected into a temporary 12-VDC configuration.

Attach the carbon-pile and clip-on ammeter as shown. Adjust the load from the carbon-pile tester to the rated performance of the alternator.

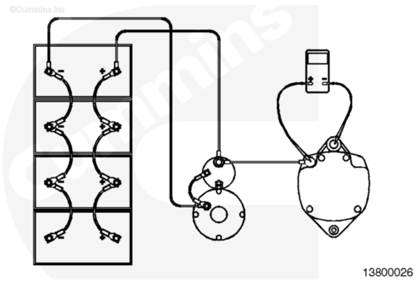
Measure the voltage drop in both the positive and negative circuits. Add these together and compare the sum to the table below.

System Voltage	Max Voltage Drop
12	0.5
24	1.0

Step B-2, Alternator Voltage Output Check

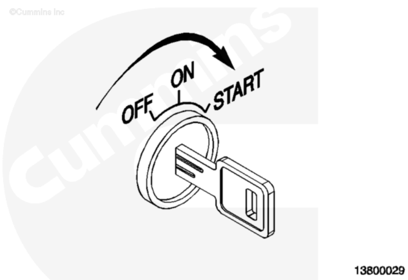
Attach the multimeter to the alternator as shown in the illustration.

With the batteries in a fully charged condition and all the accessories off, start the engine and run it at high idle. Allow time for the voltage to stabilize before taking any readings.



Measure the alternator output voltage.

System Voltage	Max Voltage Drop
12	15.5
24	31



Repair or replace the alternator or regulator if the voltage limit exceeds the value in the table.

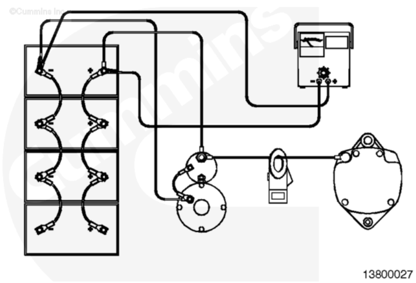
Refer to the OEM specifications for minimum voltage output.

Step B-3, Alternator Current Output Test

Connect the carbon-pile tester to the batteries in parallel.

Clamp the induction ammeter around the alternator output wire.

Note : If more than one wire is connected to the alternator output terminal, clamp the ammeter around all wires.

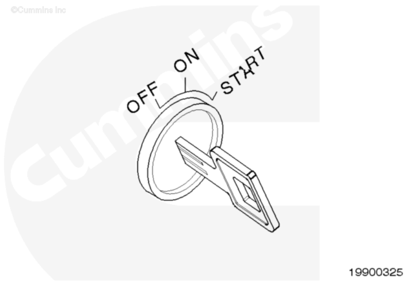


Make sure that all vehicle electrical loads are turned off.

Start the engine and operate at high idle.

Check the speed of the alternator using a digital optical tachometer. A slipping alternator drive belt can result in a low output reading. The alternator output is directly related to the speed it is turning.

Note : The alternator must be turning at approximate rated speed. Most heavy-duty alternators are rated at 5000 rpm. Check the manufacturer's specifications for the specific alternator being tested.



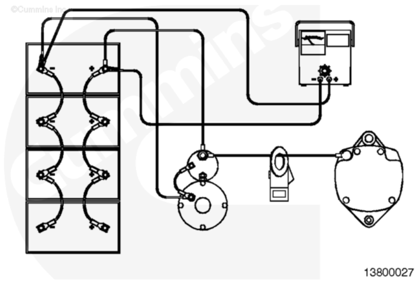
Turn on the carbon-pile tester and adjust until the ammeter reaches its highest reading. Record this value.

Turn off the carbon-pile tester and shut off the engine.

If reading on the ammeter is zero (no output), magnetize the rotor with the alternator hooked up normally. Momentarily connect a jumper lead from the battery positive (+) to the alternator relay (R) or indicator (I) terminal. This procedure applies to both negative (-) and positive (+) ground systems, and will restore the normal residual magnetism.

Repeat the test. If the output is still zero, replace the alternator.

If the alternator output is **not** within 10 percent of the rated output (stamped on the alternator case), replace the alternator.



Section 5 - Warranty Information

A. Cummins-Branded Starter and Alternator Service and Warranty

Presently, non-Cummins accessories supplied by Cummins Inc. are **not** warranted by Cummins with a few exceptions, such as package units. Cummins plants worldwide will be supplying Cummins-branded starters and alternators for engines built on or after September 15, 1996, for accessories manufactured by Delco Remy America. New Engine, New Parts, and ReCon® Warranties will apply, CAPS will **not**. The starters and alternators will have new data tags that state that this product is built for Cummins by the accessory supplier.

These products will be covered under New Engine Warranty on all applications except for the U.S./Canada RV, school bus, fire truck and crash truck where this coverage is limited to 2 years. The appropriate Legal Description will be revised to reflect this change of coverage. This warranty applies to the Cummins-branded starters and alternator **only**. If a Cummins-branded starter or alternator fails during the warranty period due to a defect in material or workmanship, Cummins, at its option, will repair/replace the starter or alternator without charge to the owner.

In order to receive warranty reimbursement for Cummins-branded starters and alternators, the Cummins-branded data tags **must** be confirmed. The Standard Repair Times for starters and alternators have been updated to reflect the use of these products. New Fail Codes for the use of Cummins-branded starters and alternators are covered in Warranty Alert 9626-A.

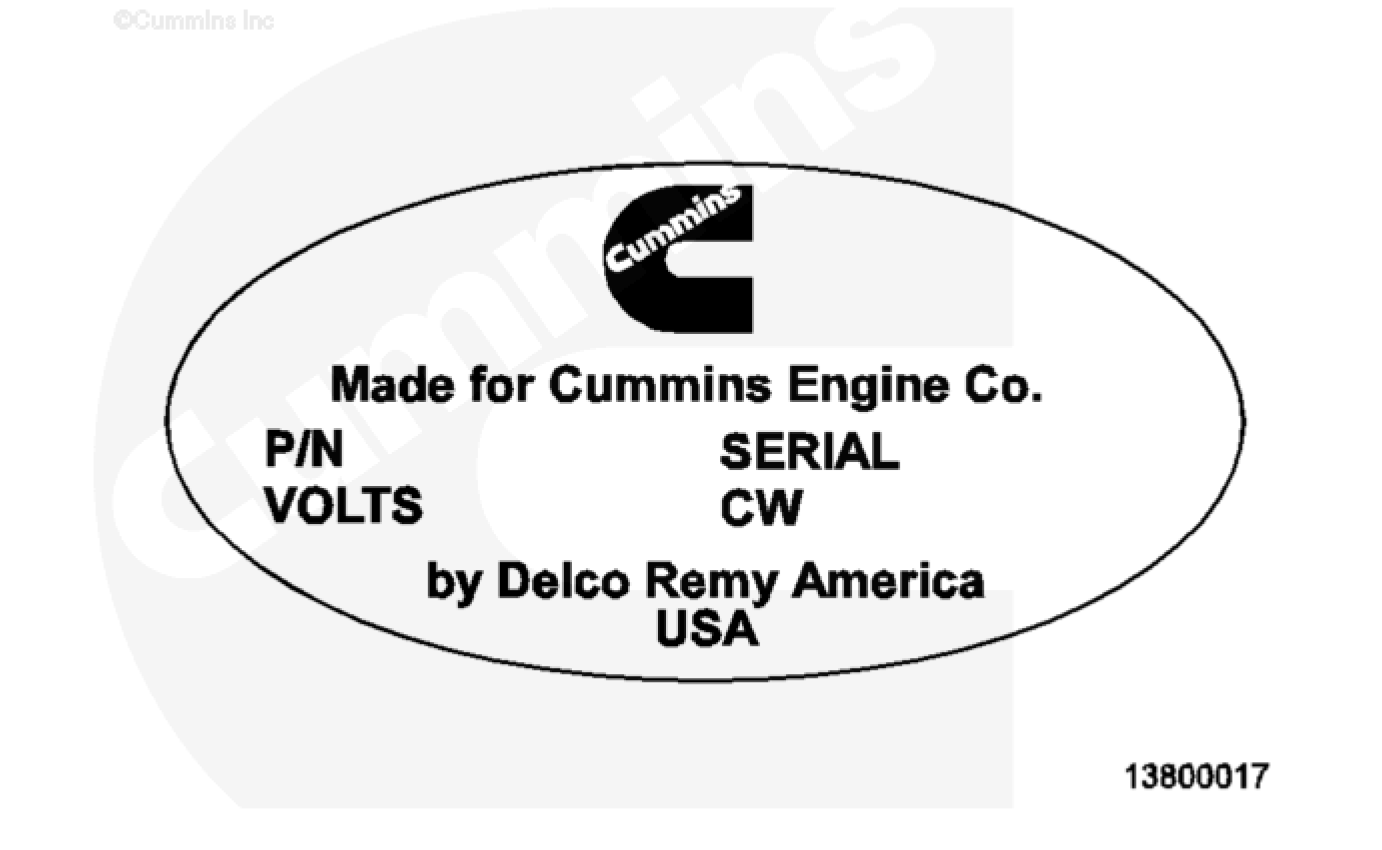


Fig. 2, Cummins and Delco Remy Tag

Note : This is the sole and exclusive warranty applicable to these products. There are no other warranties, express or implied, including no warranties of fitness for a particular purpose and merchantability. Under no circumstances will Cummins be liable for incidental or consequential damages.

Cummins-branded starters and alternators are identified by use of the following label:

B. Warranty Guidelines

The warranty of the Cummins-branded starters and alternators will be administered according to the guidelines in the Cummins Warranty Administration Manual (Cummins Form 6299).

Starter and alternator failures that are **not** considered warrantable are:

1. Customer abuse, including, but **not** limited to:
    - Handling damage
    - Stripped or overtightened threads
    - Contamination failures
    - Cranking with low system voltage
    - Excessive cranking of the starter
    - Alternator failure due to low or defective batteries.

1. Corrosion
2. Customer or OEM installation and wiring issues:
    - Magnetic switches located with the wrong orientation or a stuck switch
    - Improperly sized cables
    - Poor ground connections
    - Loose cables or connections
    - Loose customer-installed mounting bolts
    - Loose customer-installed alternator pulley
    - Milling of pinions.

1. Customer misapplication
2. No trouble found in starters and alternators.

Section 6 - Part Number Reference

This section tabulates all the Cummins-branded starters and alternators.

Cummins Inc and Delco-Remey Cross Reference				
Cummins Part Number	**Delco Part Number**	**Type**	**Voltage**	**Amperage**
135161	10479111	42MT	24	-
157579	1990389	42MT	24	-
195830	10479112	42MT	12	-
197124	1990486	42MT	12	-
3004699	10479126	50MT	24	-
3004700	10479125	50MT	24	-
3017393	10478800	50MT	24	-
3021034	10479110	42MT	12	-
3021035	10479113	42MT	12	-
3021036	10479114	42MT	24	-
3021038	10479124	50MT	24	-
3279169	1993993	42MT	24	-
3279496	1993961	42MT	12	-
3281634	10479115	42MT	24	-
3282731	10479629	28MT	24	-
3328552	10479100	42MT	24	-
3908594	10479108	37MT	24	-
3910645	10479117	42MT	12	-
3910646	10479118	42MT	24	-
3913017	10479107	37MT	12	-

Cummins Inc and Delco-Remey Cross Reference				
Cummins Part Number	Delco Part Number	Type	Voltage	Amperage
3916854	10479621	28MT	12	-
3918376	10479624	28MT	12	-
3918378	10479623	28MT	24	-
3921402	10479109	41MT	12	-
3921403	10478817	41MT	24	-
3924410	10479625	28MT	12	-
3925082	10479120	42MT	24	-
3926932	10479622	28MT	24	-
3929687	10478926	41MT	12	-
3000347	1117734	30SI	24	60
3004806	1117839	30SI	12	90
3008988	1117833	30SI	12	90
3016627	1117631	20SI	24	35
3016628	1117628	20SI	12	60
3056492	1117836	30SI	24	75
3072483	1117838	30SI	24	100
3078115	19010023	26SI	24	50
3078116	19010022	26SI	24	75
3918558	19010194	21SI	12	100
3920615	19010196	21SI	12	130
3920616	19010202	21SI	12	160
3920617	19010199	21SI	24	50
3920618	19010195	21SI	24	70
3923550	1117630	20SI	24	45
3928985	1117878	50DN	24	270
3928986	1117877	50DN	12	300

Document History

Date	Details
2016-5-10	Module Created

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